Results

Principal finding. Participation ratio (PR) and centered kernel alignment (CKA) detect some first-write timing. High timing scores can coexist with poor task accuracy, limiting claims of computation recovery.

1. Blind timing detection succeeds on average

On six held-out programs, PR and CKA achieve macro F1 of **0.853** and **0.850**, versus 0.745 for always writing. A simpler residual-change detector scores **0.876**. Activation-only predictions are frozen before schedule scoring; P11 CKA falls below its own baseline.

2. Compression exposes a gap between geometry and behavior

Across twelve tasks, target half-width uncentered SVD gives **61.62%** token accuracy, random projection 37.01%, and full rank 100%. On the held-out six, random compression reaches **0.879 CKA schedule F1** with **37.62% task accuracy** (Figure 1). The schedule remains an intended reference. Attention/MLP hidden dimensions stay fixed.

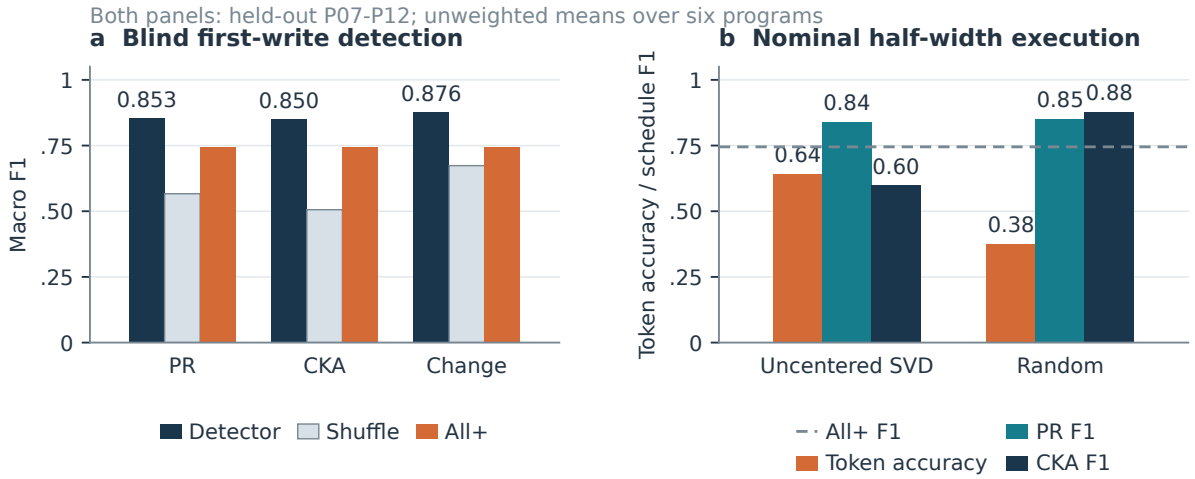


Figure 1. Unweighted means over six held-out programs, P07 to P12. Left: blind detection and baselines. Right: half-width execution. The original intended schedule is not established causal ground truth for the compressed models.

3. PCA patches provide limited causal evidence

Rank-three principal component analysis (PCA) patches match symbolic counterfactuals on **113/192 pairs (58.85%)**, versus 42.19% for equal-rank random directions and 47.92% for self patches. Conditions share 192 pairs across twelve tasks. PCA patches alter outputs in 26 pairs; variable mapping is supervised. All 60 undefined condition rows remain failures. Comparisons are descriptive.

4. Control experiments

Control	Result and interpretation
Numerical policy	Total sample-variance floor 10^{-12} reduces CKA availability mismatches from 14 to 2 across 54 transitions. Helper-state timing explains the remaining cases.
Sorting errors	Each lane fails 64/512 probes; sets are disjoint. Doubling QK-logit scale lowers maximum error from 8.52×10^{-4} to 7.15×10^{-7}; decoded accuracy stays 100%.
Rotation	All 36 executed orthogonal transforms preserve 100% output agreement. Maximum posthoc PR/CKA error is 1.42×10^{-13}; original-lane checks change.
Structural design	Nine crossed depth/branch-count variants yield rank-two realized depth/count, removing exact collinearity. No general complexity law is established.

5. Learned-model transfer

Across 36 trained transformers, per-task mean test accuracy is **95.41% to 100%**; palindrome's majority baseline is 93.94%. Final learned/compiled CKA spans 0.152 to 0.802. Learned final PR is lower for every task; internal circuits remain unknown.

Conclusion

The evidence supports partial timing recovery and modest geometry-directed intervention effects. The original fidelity comparison uses compiler-known encodings; PR depends on encoding and scale. Complete algorithm discovery, erasure, lifetimes, and semantic superposition recovery remain open. No direct InterpBench benchmark was run.

Setup. Length six; vocabulary eight. Compiled calibration/test sets contain 96/96 disjoint sequences. Learned train/validation/test splits contain 2,029/677/677 sequences, with three seeds. No experiments were rerun. Canonical evidence, e06479c.